

Covered: Inner product space isomorphisms; norm and metric isomorphisms; polarization; best approximation in finite-dimensional IPVS; existence and uniqueness for closed convex sets; orthogonality criterion for subspaces.

1 Isomorphisms over Inner Products

Let $\{u_1, \dots, u_n\}$ be an orthonormal basis of V . For $x \in V$, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $x = \sum_{i=1}^n \alpha_i u_i$. Since $\langle x, u_j \rangle = \alpha_j$, we get $x = \sum_{i=1}^n \langle x, u_i \rangle u_i$.

Consider the linear map $T : V \rightarrow \mathbb{F}^n$ as $x \mapsto (\langle x, u_1 \rangle, \langle x, u_2 \rangle, \dots, \langle x, u_n \rangle)$.

$$\begin{aligned} \text{Let } w \in \ker T &\implies Tw = 0 \\ &\iff \langle w, u_i \rangle = 0 \quad (1 \leq i \leq n) \\ &\iff \langle v, w \rangle = 0 \quad \forall v \in V \\ &\iff \langle w, w \rangle = 0 \\ &\iff w = 0 \\ &\implies \ker T = \{0\}. \end{aligned}$$

Note, that $\dim V = \dim \mathbb{F}^n = n$ and T is an injective map which implies T is an isomorphism. V and $\mathbb{F}^n$ are thus isomorphic as vector spaces.

$$\begin{aligned} \|T(x)\|^2 &= \langle T(x), T(x) \rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2 \\ \|x\|^2 &= \langle x, x \rangle = \left\langle \sum_{i=1}^n \langle x, u_i \rangle u_i, \sum_{j=1}^n \langle x, u_j \rangle u_j \right\rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2 \end{aligned}$$

T thus preserves norm. This motivates us to define the following.

Definition 1.1 (Isomorphism of Inner Product Spaces). We say that $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are isomorphic inner product spaces if there exists a bijective linear map $T : V \rightarrow W$ such that

$$\langle Tx, Ty \rangle_W = \langle x, y \rangle_V \quad \forall x, y \in V.$$

Definition 1.2. $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ are said to be isomorphic as Normed Linear Vector Spaces if there exists a bijective linear map $T : V \rightarrow W$ such that T preserves norm i.e. $\|Tv\|_W = \|v\|_V, \forall v \in V$.

Definition 1.3. We have IPVS $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ over a field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$). Then the metric induced by the inner products say d_V and d_W are defined as $d_V(x, y) = \|x - y\|_V, \forall x, y \in V$ and $d_W(x, y) = \|x - y\|_W, \forall x, y \in W$. (Here norms are those induced by the inner products)

The metric spaces thus induced are then said to be isomorphic if there exists a bijective linear map $T : V \rightarrow W$ such that we have $d_W(Tx, Ty) = d_V(x, y)$.

Note that $T : V \rightarrow \mathbb{F}^n$ as $Tx = (\langle x, u_1 \rangle \quad \langle x, u_2 \rangle \quad \dots \quad \langle x, u_n \rangle)^T$ is a bijective map that preserved inner product and norm. Also,

$$\begin{aligned} \|Tx - Ty\|^2 &= \|T(x - y)\|^2 \\ &= \sum_{i=1}^n |\langle x - y, u_i \rangle|^2 = \|x - y\|^2 \end{aligned}$$

Since, on any finite dimensional space with inner product structure, where metric is induced by norm and norm is induced by inner product ($\langle \cdot, \cdot \rangle \implies \|\cdot\| \implies d(\cdot)$) we have the following result:

Any finite dimensional vector space with inner product structure is isomorphic to $\mathbb{F}^n$ as an *Inner Product Space*, *Normed Vector Space* and as a *Metric Space*.

Hence any n -dimensional inner product space over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) is isomorphic to $\mathbb{F}^n$ as a metric space.

Corollary 1.4. Any inner product space $(V, \langle \cdot, \cdot \rangle)$ isomorphic to $\mathbb{F}^n$ is complete, i.e. every Cauchy sequence in V converges to a point of V .

Question. Does this also mean that every Inner Product on $\mathbb{R}^n$ is the same?

Answer. Consider $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_1)$ and $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle_1 = x_1 y_1 + x_2 y_2$ and $\langle (u_1, u_2), (v_1, v_2) \rangle_2 = 3u_1 v_1 + 2u_2 v_2$.

Remark 1.5. If $(V, \langle \cdot, \cdot \rangle_V)$ is isomorphic to $(W, \langle \cdot, \cdot \rangle_W)$ as vector spaces need not imply isomorphism as Inner Product Spaces. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = 2x$ is isomorphism of vector spaces. But f does not preserve Inner Product,

$$\langle f(2), f(5) \rangle = \langle 4, 10 \rangle = 40 \neq 10 = \langle 2, 5 \rangle$$

Remark 1.6. If $V \cong W$ as Inner Product, then $V \cong W$ as Normed Linear Space, further we can conclude that $V \cong W$ as Metric Spaces. Say $T : V \rightarrow W$ is a bijective linear map which is an isometry, i.e. $\langle Tx, Ty \rangle_W = \langle x, y \rangle_V$ then we have

$$\begin{aligned} \implies \|Tx\|^2 &= \langle Tx, Tx \rangle_W = \langle x, x \rangle_V = \|x\|^2 \\ \implies \|Tx - Ty\|^2 &= \|T(x - y)\|^2 = \|x - y\|^2 \end{aligned}$$

Question. If $V \cong W$ as *metric spaces* induced from inner products then are $V \cong W$ as *inner products* as well? *Answer.*

$$\begin{aligned} \|Tx - Ty\|^2 &= \|x - y\|^2 \implies \|Tx\|^2 = \|x\|^2 \quad (\text{Take } y = 0) \\ \langle Tx - Ty, Tx - Ty \rangle & \end{aligned}$$

If $\mathbb{F} = \mathbb{R}$, we have $\langle Tx, Ty \rangle = \langle x, y \rangle$

If $\mathbb{F} = \mathbb{C}$, then $\langle Tx, Ty \rangle = \overline{\langle Ty, Tx \rangle}$ and $\langle Tx, Ty \rangle + \langle Ty, Tx \rangle = \langle x, y \rangle + \langle y, x \rangle = 2a$

Say $\langle Tx, Ty \rangle = a + ib$ then $\langle Tx, Ty \rangle + \langle Ty, Tx \rangle = 2a$. Also if $\langle x, y \rangle = c + id$ then $\langle x, y \rangle + \langle y, x \rangle = 2c$. Thus $a = c$.

$$\begin{aligned} \text{Additionally, } \langle T(ix), Ty \rangle + \langle Ty, T(ix) \rangle &= \langle ix, y \rangle + \langle y, ix \rangle \\ \implies i\langle Tx, Ty \rangle - i\langle Ty, Tx \rangle &= i\langle x, y \rangle - i\langle y, x \rangle \\ \implies i(a + ib) - i(a - ib) &= i(c + id) - i(c - id) \\ \implies b &= d \end{aligned}$$

Hence we have $\langle Tx, Ty \rangle = \langle x, y \rangle$.

Remark 1.7. Alternate Proof of the above uses the polarization, For an isometry T we have $\|Tz\| = \|z\|$ for all z . If $\mathbb{F} = \mathbb{R}$, the real polarization identity

$$\langle u, v \rangle = \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 \right)$$

gives

$$\langle Tx, Ty \rangle = \frac{1}{4} \left(\|Tx + Ty\|^2 - \|Tx - Ty\|^2 \right) = \frac{1}{4} \left(\|x + y\|^2 - \|x - y\|^2 \right) = \langle x, y \rangle.$$

If $\mathbb{F} = \mathbb{C}$, use the complex polarization identity

$$\langle u, v \rangle = \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 + i\|u + iv\|^2 - i\|u - iv\|^2 \right)$$

(with inner product linear in the first variable). Applying T and using $\|Tz\| = \|z\|$ gives $\langle Tx, Ty \rangle = \langle x, y \rangle$.

2 Best Approximation

Aim: To find $\min\{\|x - w\| : w \in W\}$ (and $\arg \min\{\|x - w\| : w \in W\}$) where W is any subset of IPVS V .

Definition 2.1. Let V be an IPVS. $\emptyset \neq W \subset V$. Let $x \in V$. An element $x_0 \in W$ is said to be the *best approximation* from W to x if

$$\|x - x_0\| \leq \|x - w\|, \quad \forall w \in W$$

i.e. $\|x - x_0\| = \text{dist}(x, W)$

Question. Some natural question arises:

1. Does Best Approximation always exists for a given subset of V and $x \in V$?
2. If it exists, is it unique?
3. Is there any method to find Best Approximation?

Answer: In general best approximation may not exists. Take $V = \mathbb{R}, W = (0, 1) \cap \mathbb{Q}$. Then best approximation does not exist say for $x = 2$.

In general, if best approximation if it exists may not be unique. Consider $V = \mathbb{R}^2, W = \{(x, y) \in \mathbb{R}^2 : |x| = 1 \text{ or } |y| = 1\}$. Then at $x = (0, 0)$, the following are best approximations $(1, 0), (0, 1), (-1, 0), (0, -1)$.

For the existence, we have the following result.

Theorem 2.2. Let V be a finite-dimensional inner product space over $\mathbb{F}$. Let K be a non-empty closed convex subset of V . For every $x \in V$, there exists $x_0 \in K$ such that

$$\|x - x_0\| = \inf_{w \in K} \|x - w\|.$$

Proof. Let $d = \inf\{\|x - w\| : w \in K\}$. Choose a sequence $\{v_n\}_{n=1}^{\infty}$ in K such that $\|x - v_n\| \rightarrow d$. Then there exists a sequence $\{v_n\}_{n=1}^{\infty}$ in K such that $\|x - v_n\| \rightarrow d$. To show $\{v_n\}_{n=1}^{\infty}$ is Cauchy sequence.

Using Pythagoras Theorem, we get,

$$\begin{aligned} \left\| \frac{v_n - v_m}{2} \right\|^2 &= \left\| \frac{v_n - x}{2} - \frac{v_m - x}{2} \right\|^2 \\ &= 2 \left(\left\| \frac{v_n - x}{2} \right\|^2 + \left\| \frac{v_m - x}{2} \right\|^2 \right) - \left\| \frac{v_n - x}{2} + \frac{v_m - x}{2} \right\|^2 \\ &= \frac{1}{2} \left(\|v_n - x\|^2 + \|v_m - x\|^2 \right) - \left\| \frac{v_n + v_m}{2} - x \right\|^2 \\ &\leq \frac{1}{2} \left(\|v_n - x\|^2 + \|v_m - x\|^2 \right) - d^2 \\ &\leq \frac{1}{2} (d^2 + d^2) - d^2 = 0 \quad \text{as } n, m \rightarrow \infty. \end{aligned}$$

Thus $\{v_n\}$ is Cauchy in K . K is a closed subset of a complete metric space (since V is a finite dimensional IPVS). Hence, v_n converges to say $v \in K$.

$$\|x - v\| = \lim_{n \rightarrow \infty} \|x - v_n\| = d = \inf\{\|x - w\| : w \in K\}$$

□

Theorem 2.3 (Best Approximation Theorem). Let W be a subspace of a finite-dimensional inner product space V . For $x \in V$, an element $x_0 \in W$ is the best approximation to x from W if and only if

$$x - x_0 \perp W,$$

i.e. $\langle x - x_0, y \rangle = 0$ for all $y \in W$.

Proof. ($\implies$) *Orthogonality implies best approximation.*

Assume $(x - x_0) \perp W$. Let $y \in W$ be arbitrary. We can write:

$$x - y = (x - x_0) + (x_0 - y).$$

Since $x_0, y \in W$ and W is a subspace, $(x_0 - y) \in W$. Because $(x - x_0) \perp W$, the vectors $(x - x_0)$ and $(x_0 - y)$ are orthogonal:

$$\langle x - x_0, x_0 - y \rangle = 0.$$

By the Pythagorean theorem:

$$\|x - y\|^2 = \|(x - x_0) + (x_0 - y)\|^2 = \|x - x_0\|^2 + \|x_0 - y\|^2.$$

Since $\|x_0 - y\|^2 \geq 0$, we have:

$$\|x - y\|^2 \geq \|x - x_0\|^2 \implies \|x - y\| \geq \|x - x_0\|.$$

Thus, x_0 is a best approximation to x in W .

($\impliedby$) *Best approximation implies orthogonality.*

Assume that for all $y \in W$, $\|x - y\|^2 \geq \|x - x_0\|^2$. Expanding $\|x - y\|^2$:

$$\begin{aligned} \|x - y\|^2 &= \|(x - x_0) + (x_0 - y)\|^2 \\ &= \|x - x_0\|^2 + \|x_0 - y\|^2 + 2 \operatorname{Re} \langle x - x_0, x_0 - y \rangle. \end{aligned}$$

Subtracting $\|x - x_0\|^2$ from both sides gives:

$$\|x_0 - y\|^2 + 2 \operatorname{Re} \langle x - x_0, x_0 - y \rangle \geq 0 \quad \text{for all } y \in W.$$

Since $y \mapsto x_0 - y$ is a bijection of W onto itself, we can substitute $z = x_0 - y \in W$:

$$\|z\|^2 + 2 \operatorname{Re} \langle x - x_0, z \rangle \geq 0 \quad \text{for all } z \in W.$$

For any fixed $w \in W$ and $t \in \mathbb{R}$, replacing z with $tw \in W$ yields:

$$t^2 \|w\|^2 + 2t \operatorname{Re} \langle x - x_0, w \rangle \geq 0 \quad \text{for all } t \in \mathbb{R}.$$

For this quadratic in t to remain non-negative for all $t \in \mathbb{R}$, the coefficient of the linear term must vanish:

$$\operatorname{Re} \langle x - x_0, w \rangle = 0 \quad \text{for all } w \in W.$$

- If $\mathbb{F} = \mathbb{R}$, this immediately implies $\langle x - x_0, w \rangle = 0$ for all $w \in W$.
- If $\mathbb{F} = \mathbb{C}$, substituting $iw \in W$ into the identity gives:

$$0 = \operatorname{Re} \langle x - x_0, iw \rangle = \operatorname{Re} (-i \langle x - x_0, w \rangle) = \operatorname{Im} \langle x - x_0, w \rangle.$$

$$\text{Hence, } \langle x - x_0, w \rangle = \operatorname{Re} \langle x - x_0, w \rangle + i \operatorname{Im} \langle x - x_0, w \rangle = 0.$$

Therefore, $(x - x_0) \perp W$. □

3 Riesz Representation Theorem

Theorem 3.1 (The Riesz Representation Theorem). Let $(V, \langle \cdot, \cdot \rangle)$ be a Finite Dimensional IPVS over field $\mathbb{F}$. Let f be linear functional on V , then there exists unique vector $x_0 \in V$ such that,

$$f(x) = \langle x, x_0 \rangle$$